

Pigouvian tax or intellectual-capital royalties

Which system protects better against agentic-AI displacement? A didactic simulation report

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1 · What is being compared

System A — Pigouvian tax

A firm that replaces a worker with an AI agent pays a tax (a share τ of the labour cost it saves). The State collects that money and redistributes it to the displaced worker as an annual transfer. It is an **ex post** mechanism (it exists only after dismissal) and a **fiscal** one (it runs through the public administration, which loses 20–45% along the way to administration and deadweight).

System B — Royalties for IC transfer

The worker receives each year a share θ of the value of the algorithmic asset their knowledge helped create: a payment for the *transfer of their intellectual capital*. Rights accumulate over the career (each project generates a “vintage” that decays 15% per year) and are deposited as tokens in a custodied **wallet**, which can earn CPI, finance company projects at CPI+spread, or be withdrawn. It is an **ex ante** mechanism (it accrues while working) and a **contractual** one ($\approx 1\%$ administration cost).

Five variables to read this report

θ (theta)	The share of the algorithmic asset flow paid to the worker. This is the negotiated variable. ITEA brackets: 0.25–15% (percentiles 50–95) and 5–20% (Top 20).
τ (tau)	The Pigouvian tax rate on saved labour cost. The literature places it between 10% and 30%.
η (eta)	The income recovery rate: what fraction of net salary the worker keeps after dismissal. $\eta = 1$ means full recovery; with no mechanism, $\eta \approx 0.5$.
κ (kappa)	How many times labour cost the algorithmic asset is worth ($V = \kappa \cdot w^l$). Calibrated with profit per employee of the world's 2,000 largest firms: 1.3 (retail) to 2.2 (banking).
θ^* (frontier)	The indifference θ : below it the worker prefers the tax; above it, the royalty. The whole report revolves around where this frontier lies.

2 · The full picture: from age 35 to 65

The figure shows a Spanish financial analyst (ITEA percentile 94, €77,500 gross) displaced at 55. The grey line is life without automation. Blue is the Pigouvian system: nothing changes until dismissal; afterwards the fiscal transfer restores part of the income. Green is the cumulative royalty system: for 20 years IC rights accumulate in the wallet (shaded area), which reaches dismissal with about €270,000 — pre-funded self-insurance that, together with the decaying royalties, sustains income better than the transfer. The key

is not that the royalty pays more: it pays **when the knowledge transfer happens**, not when the dismissal does.

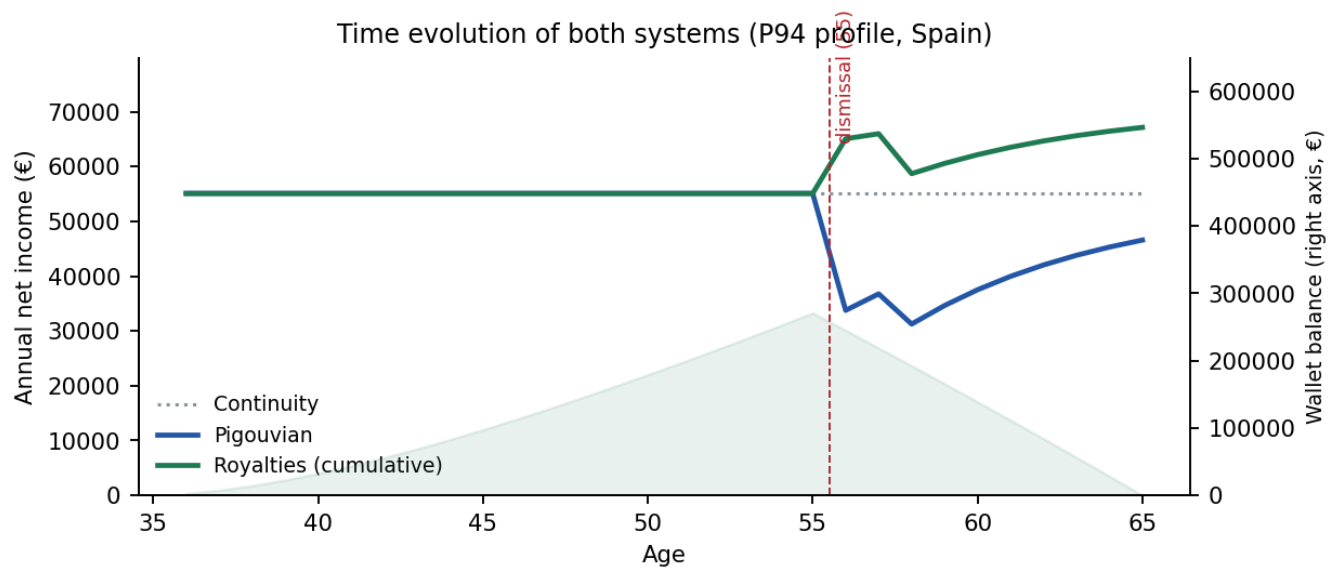


Fig. 1 — Worker income and wallet, ages 35→65.

3 • The θ^* frontier: how much royalty it takes to beat the tax

For every country and tax configuration we compute the θ^* above which the worker is better off with royalties. The frontier turns out narrow and low: between 1.5% and 7.7% across the entire plausible space. Read it like this: with a 20% tax and 35% fiscal inefficiency, a θ of 3.5–4.1% suffices for the royalty to dominate — and the ITEA brackets allow negotiating up to 15–20%. The tax only wins if the royalty is calibrated very low, or if the fiscal channel were exceptionally efficient.

θ^* (cumulative, dismissal at 55)	$\tau = 10\%$	$\tau = 20\%$	$\tau = 30\%$
Fiscal inefficiency 20%	2.2 – 2.6%	4.4 – 5.1%	6.5 – 7.7%
Fiscal inefficiency 35%	1.8 – 2.1%	3.5 – 4.1%	5.3 – 6.2%
Fiscal inefficiency 45%	1.5 – 1.8%	3.0 – 3.5%	4.5 – 5.3%

Minimum θ for royalties to dominate the tax (worker, dismissal at 55)

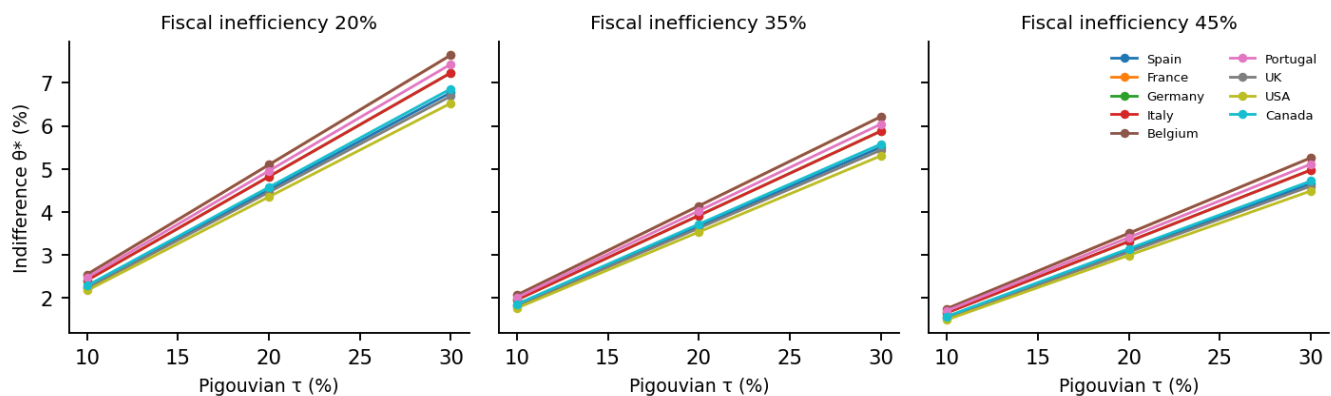


Fig. 2 — θ^* frontier by country, τ and inefficiency.

4 • The central finding: age reverses the ranking

Under the cumulative architecture, the indifference θ^* **falls** with dismissal age: a worker displaced at 60 arrives with 25 years of accrued rights and needs only $\theta = 1.5\%$; one displaced at 45 has barely accumulated and needs 8–9%. Under the flat architecture the opposite happens. The didactic conclusion: **the two systems do not compete — they complement each other across ages**. The tax is the natural insurance of the young displaced worker (a transition problem); the cumulative royalty is the senior's (an expropriation-of-decades problem). A hybrid design — a fiscal transfer decreasing with tenure plus a royalty increasing with it — would dominate both, crossing around ages 48–52.

Displacement age reverses the ranking

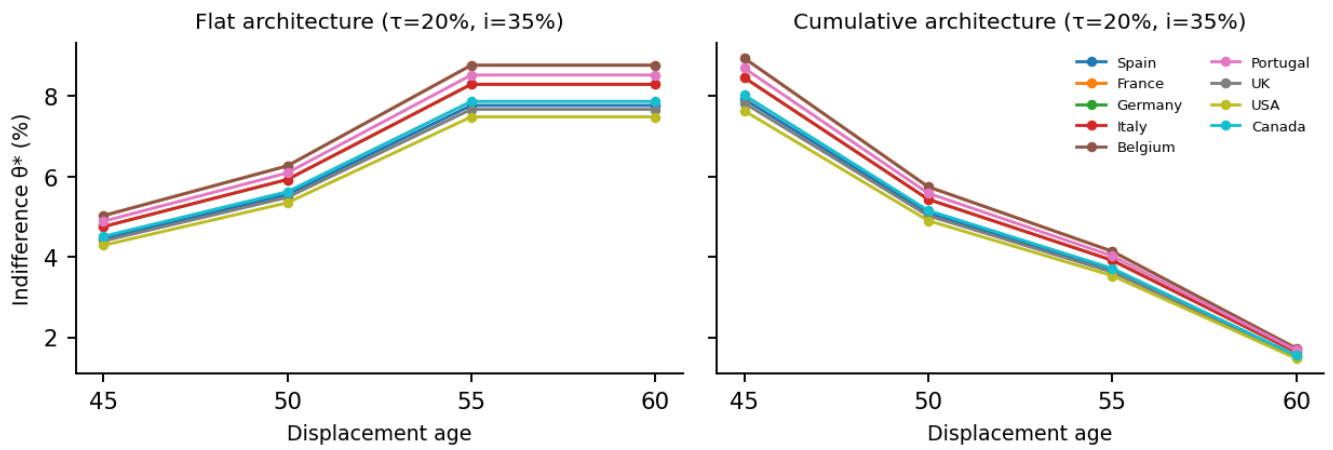


Fig. 3 — θ^* by dismissal age and architecture.

5 · Each country's tax structure moves the frontier

Three national parameters decide how much each system delivers. **The tax wedge** (how much of labour cost never reaches the worker): where it is high (Belgium 54%, Germany, France) the tax base is huge — the same τ collects 40% more in France than in the US — but so is the fiscal cost of each dismissal. **Baseline protection η_0** : in the Anglo bloc, public benefits replace less than half the salary ($\eta_0 = 0.46$), so any mechanism adds more value — and the royalty wins earlier precisely where it is most needed. **Capital taxation**: royalties are taxed as savings income (18% in the US, 30% in France and Belgium), and every point raises the required θ . Result: the tax resists best in the high-wedge continental bloc; the royalty sweeps the Anglo bloc. For the State and the taxpayer there is no contest: the royalty costs 26–39% less in all nine countries.

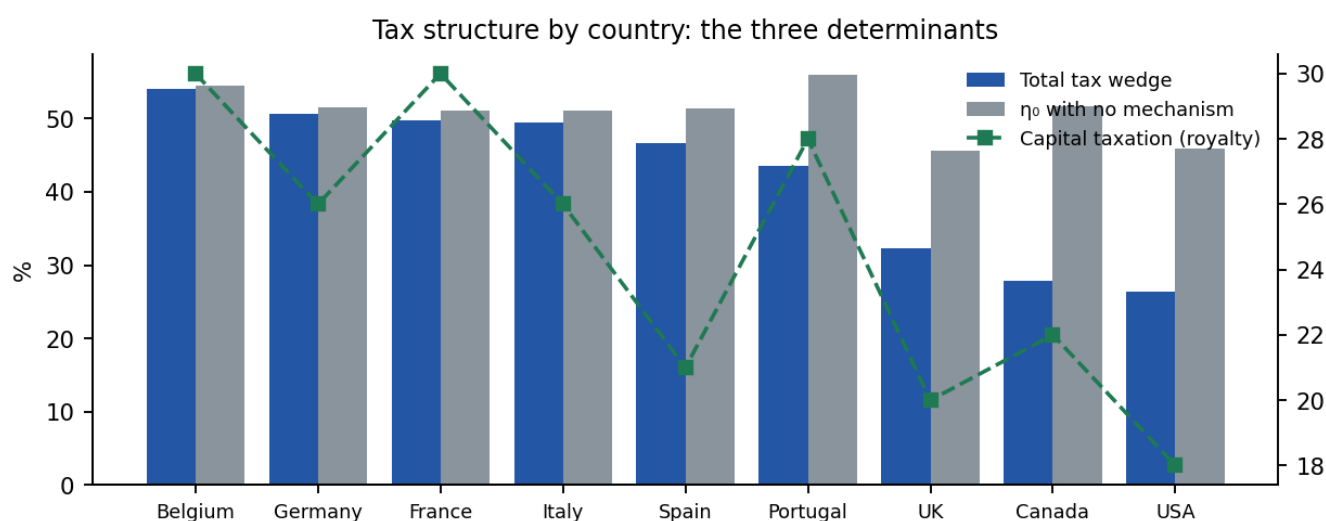


Fig. 4 — Tax wedge, η_0 and capital taxation by country.

6 · The firm: it pays earlier, but a bargaining zone exists

The firm's perspective is the mirror image. With similar annual flows (10% royalty: €14,620/yr; 20% tax: €16,244/yr), the present value of the royalty **doubles** that of the tax (€152,700 vs €76,700 per worker) — because the tax is paid late and the royalty from year one. Two buffers make it contractible. First, **internal financing**: the wallet is a liability the firm remunerates at CPI+spread, below its cost of debt; for 1,000 workers the pool reaches €180m and recovers 16% of the cost. Second, and decisive: crossing the firm's frontier (max acceptable θ : 5.0% at $\tau=20\%$) with the worker's (min θ : 3.7%) reveals a **contracting zone** — θ between 3.7% and 5.0% — where both prefer the royalty to the tax. And the zone widens with τ : **the tax is not the alternative to the royalty; it is the credible threat that makes it negotiable.**

Firm cost and the contracting zone

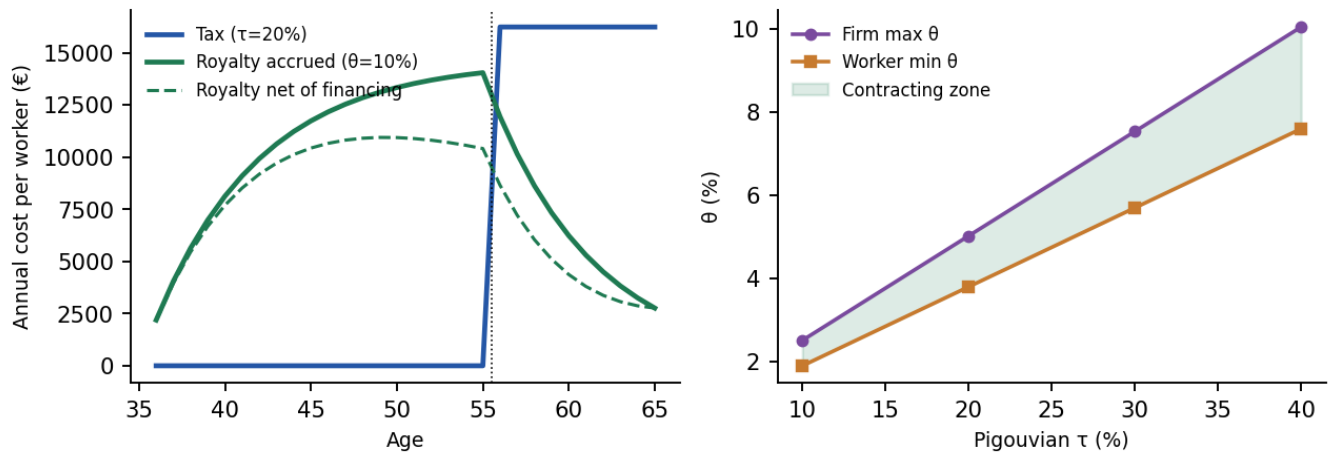


Fig. 5 — Annual firm cost and the contracting zone.

7 · Externalities: what each system corrects and what risks it creates

What the royalty corrects better

- The demand externality: it sustains income and consumption without losing 20–45% in the fiscal channel.
- Evaporated revenue: the State taxes the flow for 30 years, not only after dismissal.
- Knowledge hoarding: paying for IC transfer removes the incentive to hide it (the tax incentivises the opposite).
- The political cycle: it is contractual; it survives annual budgets.

The risks the royalty must manage

- “Enron” risk concentration: job, savings and credit in the same restructuring firm → mandatory guarantee fund.
- Acceleration: with rights already accrued, dismissal becomes marginally cheaper → accrual conditional on continued employment.
- Disintermediation: a national pool of €20–30bn outside banking supervision requires regulatory transparency.
- Under-declaration of V: requires auditability (usage metadata, Shapley attribution).

8 · Conclusion: when each system is optimal

The Pigouvian tax is optimal when...

- θ negotiated below 2–4% (bracket floor)
- The displaced worker is young or low-tenure (< 12 years of IC)
- The occupation is below the 50th ITEA percentile (the only universal instrument)
- High tax-wedge, high capital-tax country (Belgium, France)
- The fiscal channel is exceptionally efficient ($i \leq 20\%$) with credible earmarking

The cumulative royalty is optimal when...

- $\theta \geq 4\text{--}5\%$ (any mid-to-high ITEA bracket)
- Accumulated career ≥ 15 years; senior displaced workers ($\geq 50\text{--}55$)
- Weak public protection country (US, UK, Canada)
- Moderate capital taxation ($\leq 22\%$: Spain, Anglo bloc)
- Always, from the State's and the taxpayer's perspective (26–39% cheaper)

The design recommendation that follows from the simulation: legislate the tax as a subsidiary regime ($\tau \geq 20\%$) and allow opting out into cumulative royalties negotiated within the contracting zone, with a mandatory insolvency guarantee, a fraction of the deposit allocated to pension contributions, and the internal-financing spread as the distribution variable in collective bargaining.

Quick glossary

w^t	Total labour cost: gross salary + employer social contributions. The tax base and the reference for V.
V	Annual flow of the algorithmic asset attributable to the worker: $V = \kappa \cdot w^t$.
Vintage	The “harvest” of rights generated by one year's projects; decays 15% per year (mean life ≈ 6.7 years).
Wallet	Custodied personal account where rights accumulate as tokens; three destinations: CPI deposit, internal financing (CPI+spread) or withdrawal.
i	Fiscal channel inefficiency: the share of revenue lost to administration and deadweight (20–45%).

η_0	Income recovery with no mechanism: what public benefits replace on their own (≈ 0.46 – 0.56 by country).
Contracting zone	θ interval where worker and firm both prefer the royalty to the tax: [3.7%, 5.0%] at $\tau = 20\%$.

Methodological notes: approximate parametric national taxation (2026); simplified unemployment systems (capped two-year benefit + assistance); reference profile P94/ $\kappa=1.8$; real terms with 3% discounting; ITEA percentiles in the example are illustrative. The lifetime comparison includes the wallet pre-funding, paid by the firm during the career: the royalty is not cheaper in total cost — it is better in timing. Results reproducible with the ITEA Simulator engine (fronteras.py, modelo.py, acumulativo.py).